

The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines transformed public health by eliminating or drastically reducing deadly diseases over the past two centuries.

1796–2020

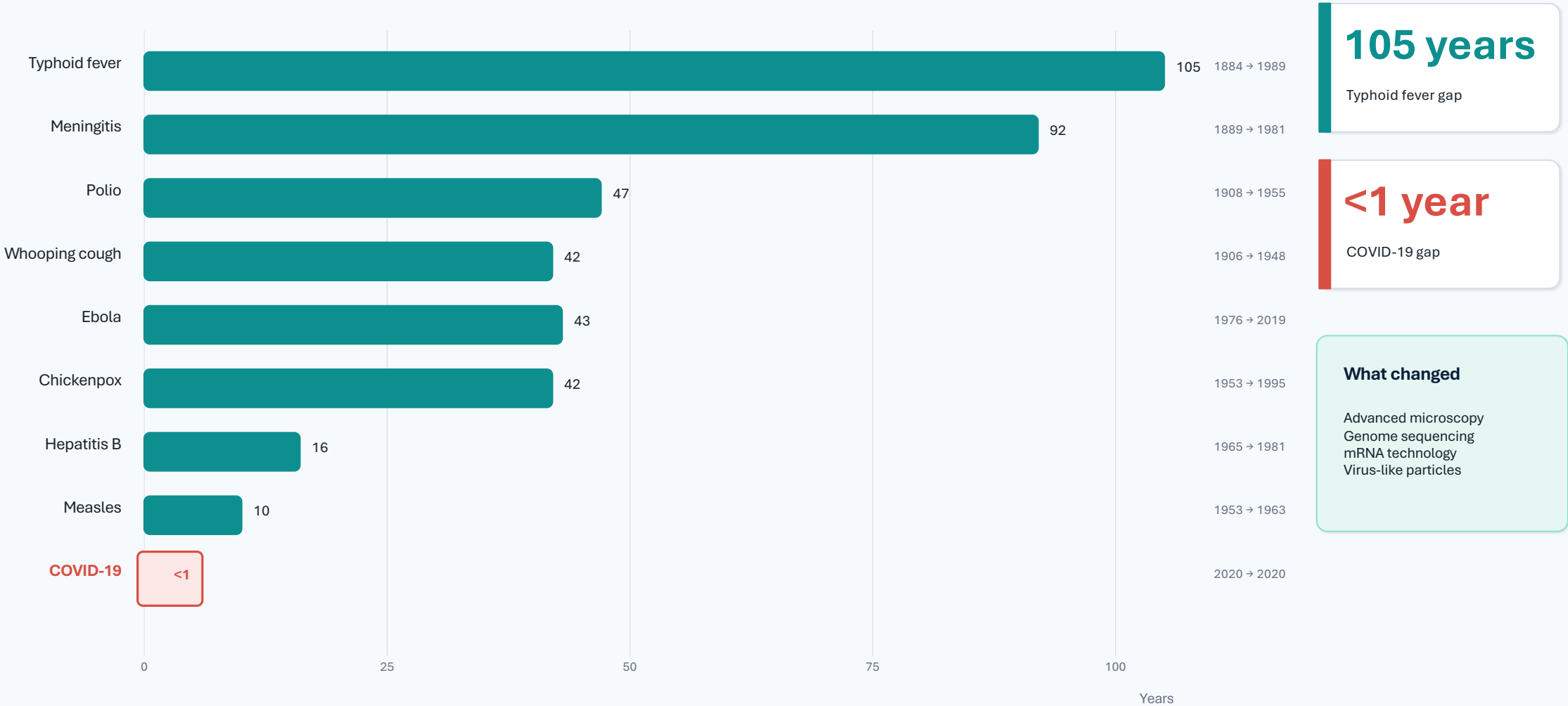
PUBLIC HEALTH + POLICY

Source: Our World in Data



From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time span from pathogen identification to vaccine licensure



COVID-19's record timeline is visually compressed because the licensure gap was under one year.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre- and post-vaccine annual burden per million people, with remaining burden visible

Disease	Pre-vaccine cases	Post-vaccine cases	Case reduction	Pre-vaccine deaths	Death reduction	Status
Diphtheria	158	0	100%	13.7	100%	100% reduction
Smallpox	250	0	100%	2.9	100%	100% reduction
Acute polio	141	0	100%	10	100%	100% reduction
Measles	3,044	0.2	99.99%	2.5	100%	reduced
Rubella	242	0.04	99.98%	0.09	100%	reduced
Pertussis	1,534	52	96.6%	30.8	99.7%	reduced
Hepatitis A	465	51	89%	0.5	88.7%	reduced
Acute hepatitis B	273	44	83.9%	1	83.6%	reduced
Pneumococcal	233	139	40.5%	24	31.3%	largest gap

100%

Case and death elimination for diphtheria, smallpox, acute polio

40.5%

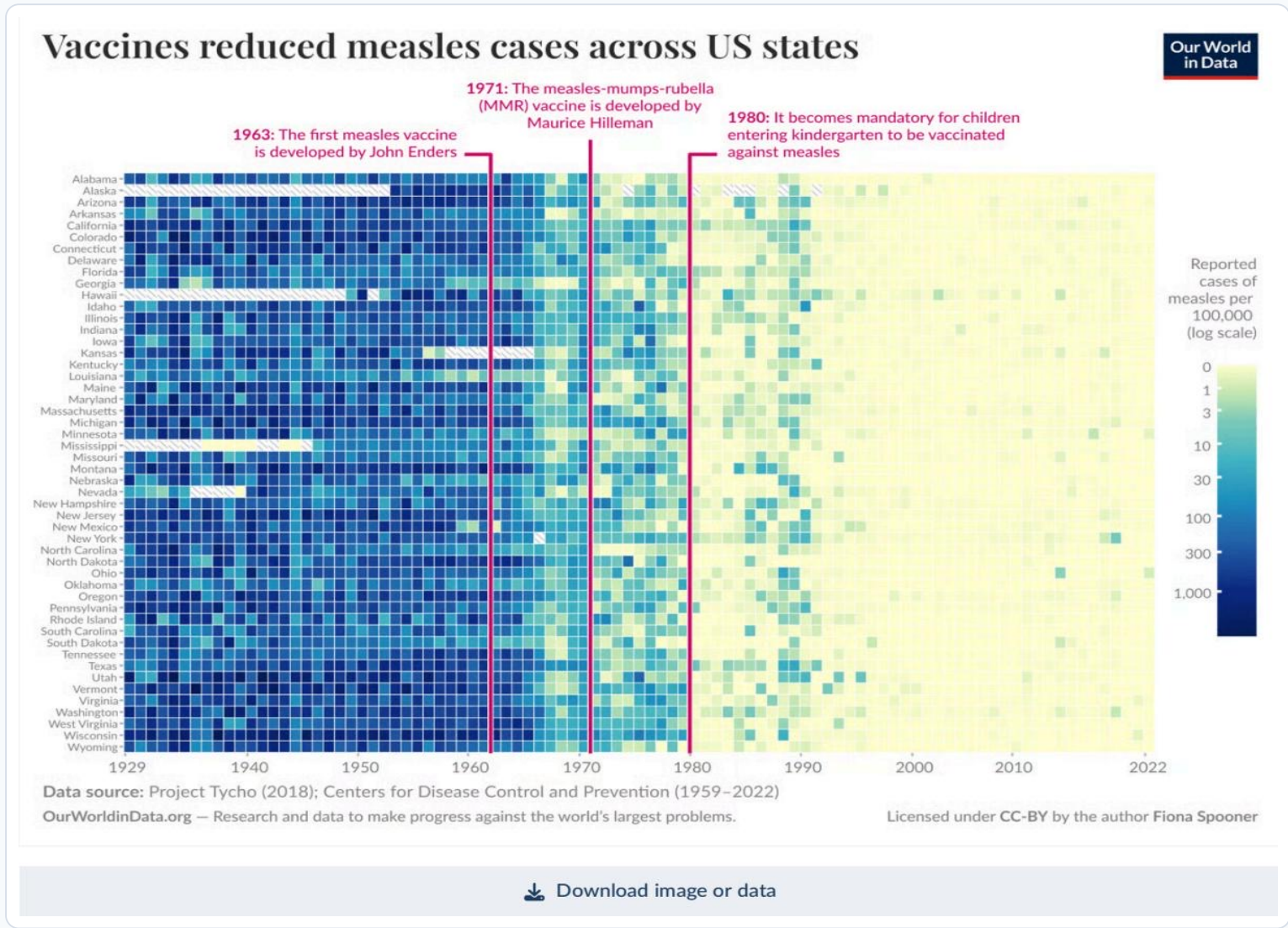
Pneumococcal case reduction remains lowest

99.99%

Measles case reduction; deaths reduced 100%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

State-level heatmap, 1929–2022, annotated with vaccine policy milestones



Reading the heatmap

- 1963** First measles vaccine developed by John Enders
- 1971** MMR vaccine developed by Maurice Hilleman
- 1980** Vaccination became mandatory for children entering kindergarten

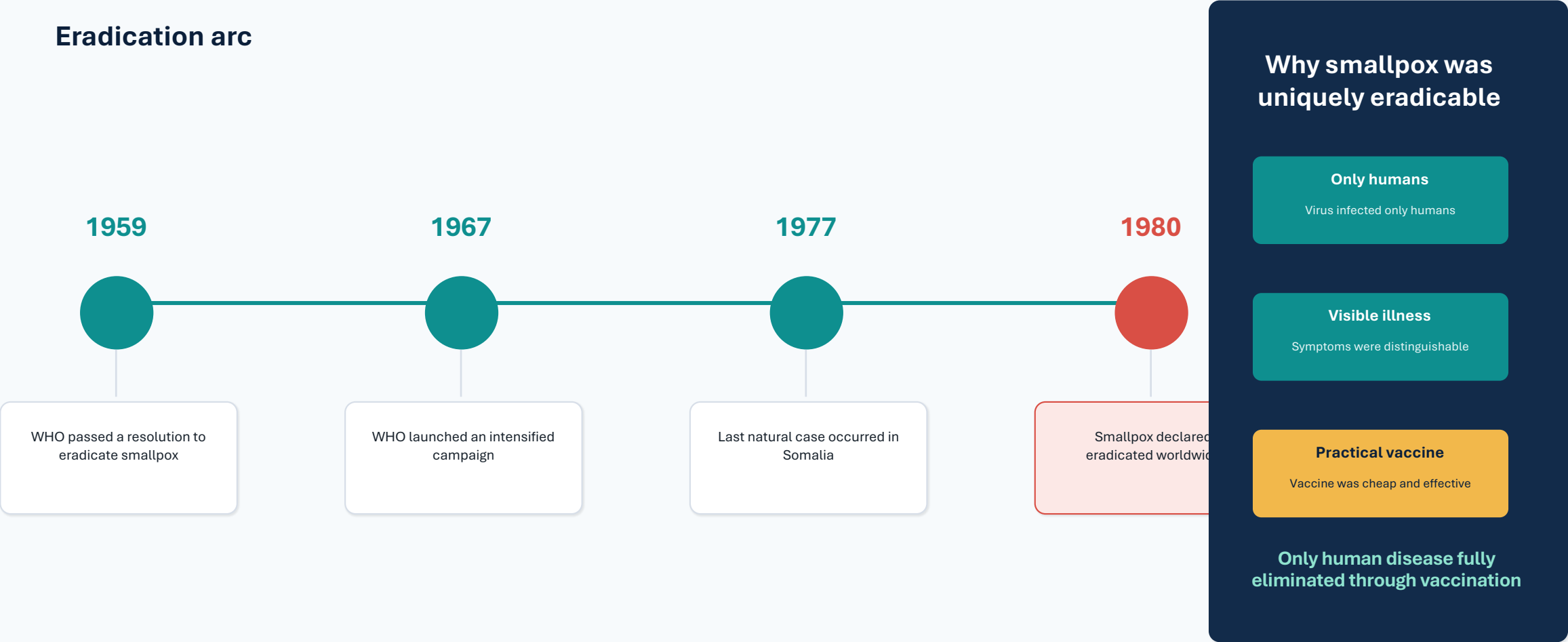
Before 1963 Nearly every state had high incidence: 100–1,000+ cases per 100,000.

After 1980: near zero across all states, with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination

A global campaign progressed from resolution to last natural case to certification

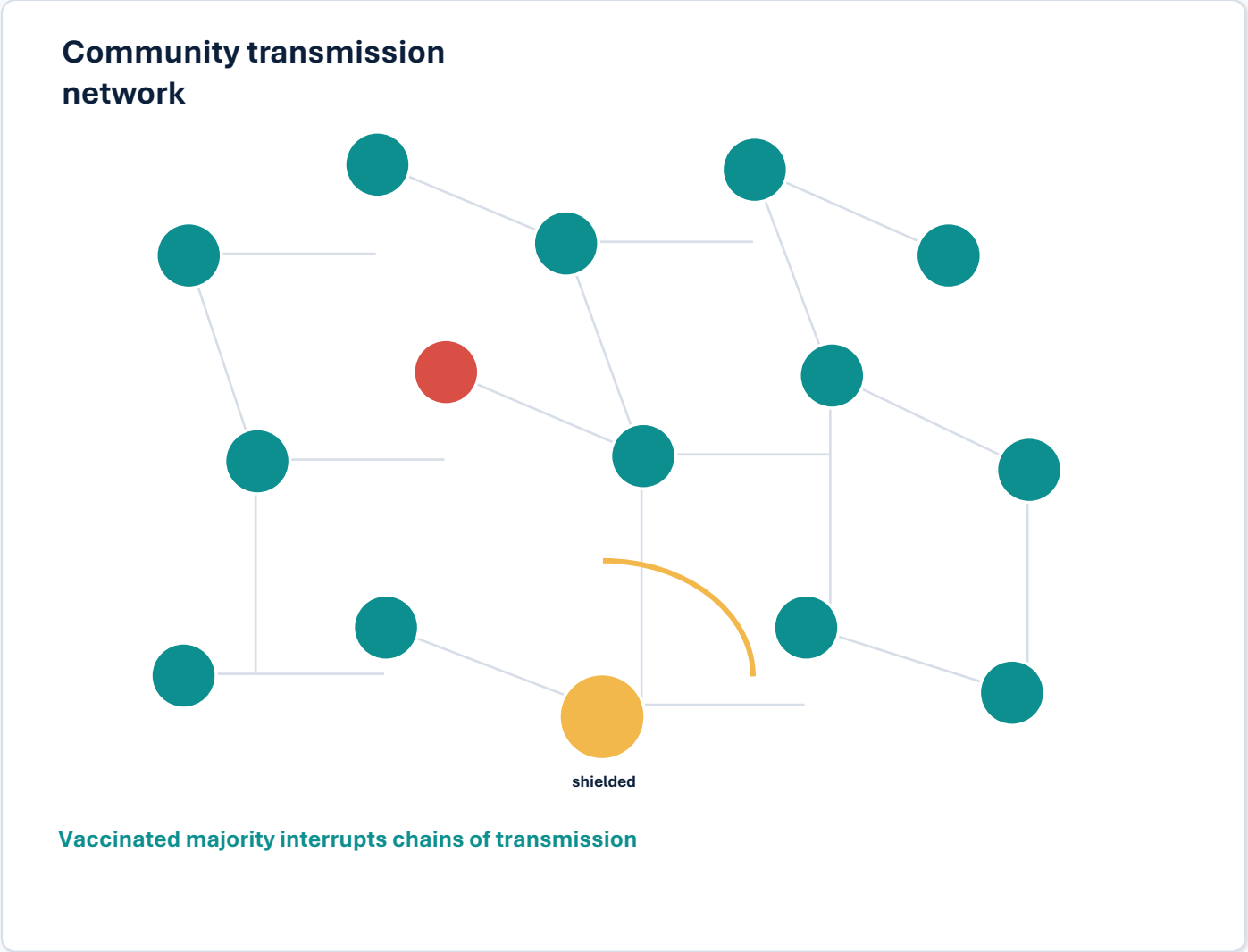
Eradication arc



No country counts are shown because the provided brief contains milestones, not endemic-country time series data.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively



1

Individual protection

Vaccination reduces susceptible hosts.

2

Community effect

Pathogens cannot spread effectively when enough people are vaccinated.

3

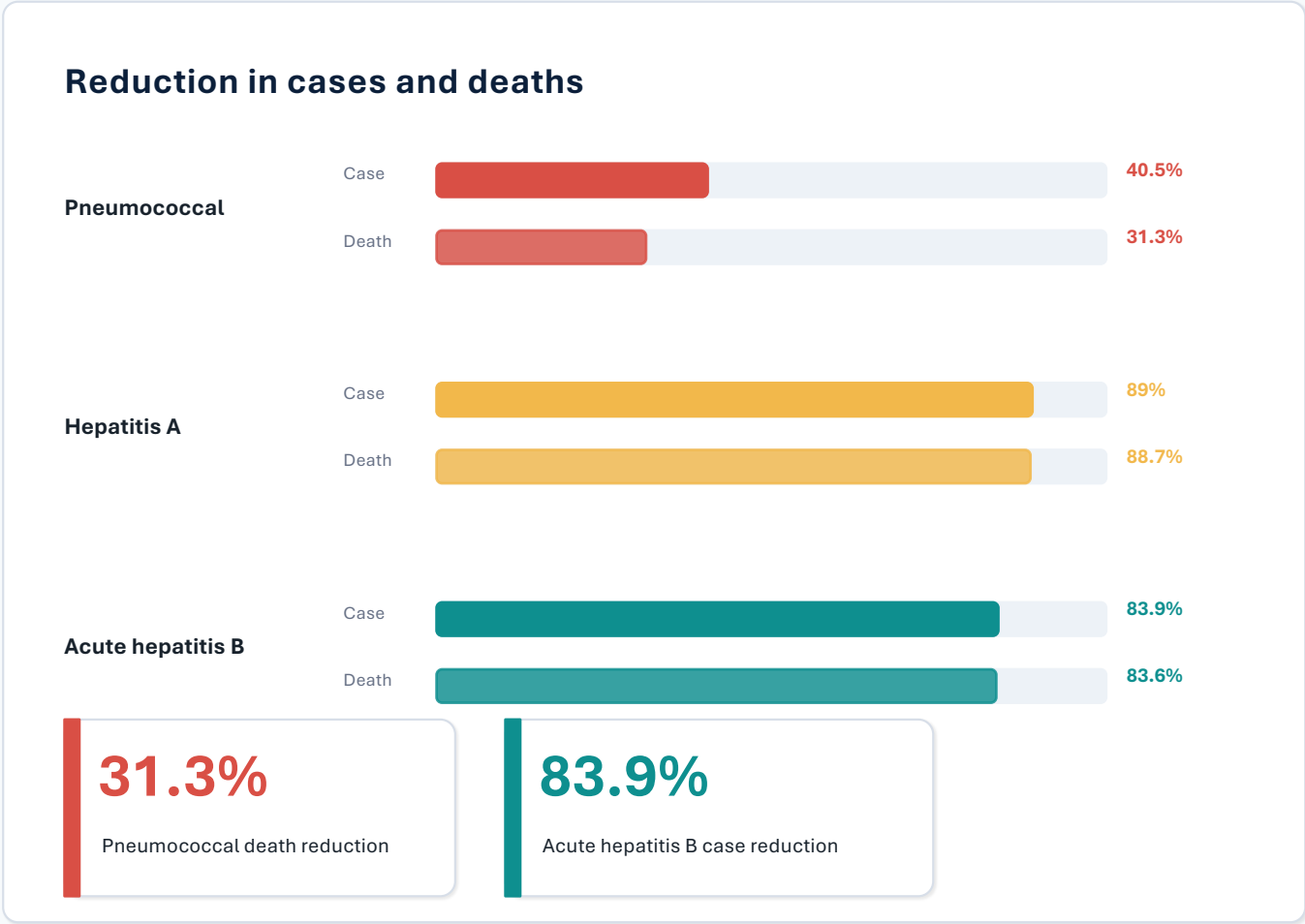
Vulnerable shielded

Newborns, immunocompromised individuals, and cancer patients on chemotherapy benefit from lower exposure.

Legend: teal = vaccinated, coral = susceptible, gold = medically vulnerable

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete control signals access, supply, trust, and vaccine-modernization priorities



Why gaps remain

Access gaps

Not everyone has access to vaccines globally

Supply shortages

Shortages put children at risk

Trust barriers

Poor scientific understanding can reduce trust and uptake

Priority: improve access, coverage, and newer vaccines where burden persists.

Key Takeaways

Vaccines have eliminated deadly diseases — but the job is not done.

1

Elimination is possible

Diphtheria, smallpox, and acute polio reached 100% case and death elimination in the US.

2

Innovation accelerated

Development timelines collapsed from over 100 years to under 1 year.

3

Mandates changed measles

Cases dropped to near zero across all 50 US states after the 1980 kindergarten mandate.

4

Community protection matters

Herd immunity shields newborns, immunocompromised individuals, and cancer patients on chemotherapy.

5

Gaps require action

Pneumococcal, hepatitis A, and acute hepatitis B reductions remain incomplete.

Call to action

Strengthen global systems for vaccine access, coverage, supply resilience, and public trust.